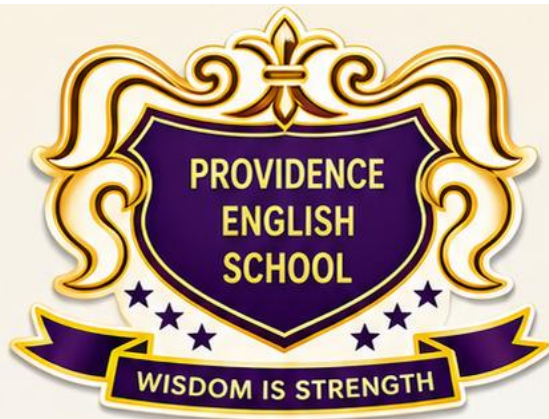




# PRIMARY COMPUTING

➤ Building skills. Creating ideas. Shaping the future. ➤

YEAR  
1

when  clicked  
move 10 steps  
say Hello!  
repeat 4  
draw 

</>

EXPLORE  
LEARN  
CREATE

THINK

DESIGN

CREATE

SHARE

# Unit1: Understanding Computer Systems

## Lesson 1: Different Computer Systems

### Learning Objective

**1CS.01 Know that there are a range of computer systems with different functions.**

**You will learn:**

**That computers can be used for different things.**

**That computer can run different programs.**

### What is a computer?

Today we are going to learn about computers.



A **computer** is a machine that helps us do many jobs. It cannot think for itself; it only follows instructions.

We use computers to learn, play games, watch videos, draw pictures, and talk to people."

### Different Computer Systems

#### Desktop Computer

- Stays on a desk.
- Used at school or home.
- Good for learning and working.





## Laptop

- Easy to carry.
- Can be used anywhere.
- Good for school and home.



## Smartphone

- Fits in your hand.
- Used for calling, taking photos, and apps.



## Tablet

- Has a touch screen.
- Used for games, reading, and learning.



# Parts of a Computer

These are the main parts of a computer.



In the UAE, we use computers every day to **learn, work** and **stay connected**!



### 1. Monitor (Screen)

It shows pictures, words and videos.

In the UAE, we watch videos and online lessons, and see information on the screen.



### 2. System Unit (CPU)

It is the brain of the computer. It does all the work.

Many offices and schools in the UAE use computers every day.



### 3. Keyboard

We use it to type letters, numbers and words.

Students in the UAE use keyboards to write assignments and projects.



### 4. Mouse

We use it to move the pointer, click and select things.

We use a mouse to browse the internet, book tickets and shop online in the UAE.



### 5. Speakers

They let us hear sounds and music.

We listen to Emirati songs, stories and educational videos.



### 6. Headphones

We use them to listen to sounds quietly.

We use headphones in class or at home without disturbing others.



### 7. Microphone

We use it to speak and record our voice.

We use microphones for online classes, presentations and recording.



### 8. Printer

It prints our work on paper.

We print homework, tickets, forms and important papers.



### Did You Know? (UAE)

The UAE is a leader in technology! We use computers in many important places like schools, hospitals, airports, shops, banks and government offices.



All parts work together to help us **learn, play, create** and **communicate**!



Computers help make life **easier** and our future **brighter** in the UAE!







# Where Can We Use Computers?



**We can use computers in many places!**



In the UAE, computers help us **learn, work, stay safe** and build a **better future!**

## At Home



**We can learn, play  
and have fun.**

 In the UAE, many families use computers to watch educational videos and stay connected with loved ones.

## At School



**We can learn, do our  
work and research.**

 Students in the UAE use computers to research, do projects and learn new things.

## At Work / Office



**We can write, send emails  
and create documents.**

 In the UAE, many jobs use computers for emails, meetings, reports and organising work.

## At the Library



**We can find information  
and read e-books.**

 In the UAE, computers help us search for information and read digital books.

## In the Hospital



**Doctors use computers to  
help patients and keep records.**

 Hospitals in the UAE use computers to store patient records, book appointments and do tests.

## At the Store



**Stores use computers for  
billing and stock.**

 Shops in the UAE use computers to scan items, print bills and check stock and prices.



**Computers help us learn, work, communicate and do many important things every day!**  
**They help us build a **smart, safe** and **happy** future in the **UAE!****



# Practise

## Match the Input Device

Draw a line to match the picture with its name.

Keyboard



Mouse



Microphone



Webcam



Scanner



## Match the Device to Its Function

### Device



Keyboard



Mouse



Microphone



Webcam



Scanner

### Function



Speak to the computer



Type letters and numbers



Click and move



Scan pictures



Take photos and videos

## Fill in the Blanks

choose the correct word from the Word Bank.

**School | Hospital | Store | Home**

1. We can play and have fun on the computer at \_\_\_\_\_.
2. Doctors use computers to help patients in the \_\_\_\_\_.
3. We can learn and do research with computers at \_\_\_\_\_.
4. People use computers for billing and stock at the \_\_\_\_\_.

## Name the parts of the computers

**Keyboard      Mouse      Microphone      Webcam**  
**Monitor      CPU      Speakers      printer**





## Lesson 2: Programs and Computers

### Learning Objective

1CS.02 Know that computers can run many different programs.

You will learn:

Computers can do many different things.

We use computers:

Computers can do many different things:



### Computers Help Us In Many Ways!

In the UAE, we use computers every day!





#### Draw pictures.



 Kids in the UAE draw beautiful pictures of landmarks like the Burj Khalifa and Sheikh Zayed Grand Mosque.



#### Solve maths.



 Students in the UAE use computers to practise maths and solve problems in school.



#### Listen to music.



 We can listen to Emirati songs and enjoy music by talented artists from the UAE.



#### Play games.



 Children play fun and educational games that help them learn while celebrating UAE culture.



#### Can talk with friends.



 We can have video calls with family and friends, even if they are in another Emirate or country.



#### Sending emails.



 We can send emails to our teachers and family, share our work and stay connected.



Computers are amazing! They help us learn, create, communicate and have fun! In the UAE, computers help us build a **smart** and **happy** future!





# Computers Can Run Many Different Programs

A program is a set of instructions that tells the computer what to do. Computers can run many different programs to help us every day!

Paint Program



We can draw pictures.

Calculator Program



We can solve maths problems.

Music Program



We can listen to music.

Games Program



We can play games.

Email Program



We can send emails.

Chat Program



We can talk with our friends.

LAPTOP



A laptop can run many different programs at the same time!

TABLET



A tablet can run many different programs too!



Different devices can run the same kinds of programs!

SMART WATCH



Even a smart watch can run useful programs!



So, laptops, tablets and smart watches are all computers that can run many different programs to help us learn, create, communicate and have fun!

A program is a set of instructions that tells the computer what to do. A computer can run many different programs.

Paint Program



We can draw pictures.

Games Program



We can play games.

One computer...  
Many programs!



Calculator Program



We can solve maths problems.

Music Program



We can listen to music.

Email Program



We can send emails.

Chat Program



We can talk with our friends.

Different programs help us do different things!



So, one computer can run many different programs to help us learn, create, communicate and have fun!



# What is my Job?

Draw a line to match each computer part to its special job.

## Screen



**I help you type  
letters and words**

## Mouse



**I show pictures  
and video**

## Keyboard



**I click and move  
things on screen**

# Name the Computer

Look at the shape of the computer. Write the correct word under each picture.

Laptop

Tablet

Desktop

Smart Watch



- Circle the things a computer can do.

Play a game

Wash the dishes

Show a picture

Type a story

- Join each action to the matching picture.

**Play games**



**Draw pictures**



**Watch a video**



- What can a computer help us do? Tick the **three** correct pictures.



• ☐ Solve Maths

• ☐ Eat lunch

• ☐ Play games

• ☐ Talk to friends



## Lesson 3: Input Devices

### Learning Objective

1CS.03 Know that information and data can be input to computers in many different ways.

You will learn:

Know that information and data can be input to computers in many different ways.



Input devices are used to **send information to the computer**. They help us give instructions and data to the computer.

## INPUT DEVICES AND THEIR FUNCTIONS



Input devices are used to **send information to the computer**. They help us give instructions and data to the computer.



In the UAE, we use technology every day to learn, work, travel and build a better future! ❤️

| Input Device | Picture | Function                                                    | Example in the UAE                                                                                                        |
|--------------|---------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Keyboard     |         | We use it to type letters and numbers.                      | Students in the UAE use keyboards in school to write, research and complete their tasks.                                  |
| Mouse        |         | We use it to move the pointer, click and select things.     | People in the UAE use a mouse to book tickets, shop online and navigate smart websites.                                   |
| Microphone   |         | We use it to speak to the computer.                         | We use microphones in the UAE to attend online classes, record voice messages and speak to virtual assistants like Alexa. |
| Webcam       |         | We use it to take pictures and videos.                      | Webcams help families in the UAE stay connected through video calls with loved ones near and far.                         |
| Scanner      |         | We use it to scan pictures and documents into the computer. | Scanners are used in UAE offices, schools and hospitals to scan documents, forms and important papers.                    |



Input devices help us give instructions to the computer.

In the UAE, technology helps us learn, work and make our lives easier every day! ❤️ 🇦🇪

**Activity:** Circle the input devices.



### Did You Know? (UAE)

The UAE is a leader in smart technology! We use computers and input devices in amazing places like schools, airports, hospitals and space centres!



## Tick (✓) the Input Devices

- ☐  Mouse
- ☐  Printer
- ☐  Keyboard
- ☐  Speakers
- ☐  Microphone

|                                                         |
|---------------------------------------------------------|
| <b>Word Bank</b>                                        |
| <b>Keyboard / Mouse / Microphone / Webcam / Scanner</b> |

1. We use a \_\_\_\_\_ to type letters and numbers.
2. We use a \_\_\_\_\_ to move the pointer, click and select things.
3. We use a \_\_\_\_\_ to speak to the computer.
4. We use a \_\_\_\_\_ to take pictures and videos.
5. We use a \_\_\_\_\_ to scan pictures and documents into the computer.



# Lesson 4: Output Devices

## Learning Objective

1CS.04 Know that computers can output information in many different ways.

Year 1

## Lesson 4: Output Devices



Computers can output information in many different ways.



### Learning Objective

1CS.04

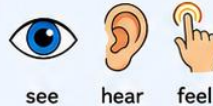
Know that computers can output information in many different ways.

In the UAE, computers help us learn, work, travel and live in smart ways!



### What is Output?

Output is the information that comes out of a computer. We can see it, hear it, or even feel it.



see hear feel



In the UAE, computers show information on screens, in shops, schools, airports and many other places!



### Key Idea

Computers do not think or make decisions by themselves. They take our input (like pressing keys or clicking) and give us output (the results).



### Examples of Output Devices

These devices show, play or give information from the computer.

#### 1 Monitor



A monitor shows us pictures, words, and videos.

##### Example:

Watching a lesson on the screen.

Used in UAE schools for smart learning.



#### 2 Printer



A printer makes hard copies on paper.

##### Example:

Printing your homework or a picture.

Used in offices and government centres in the UAE.



#### 3 Speakers



Speakers let us hear sounds and music.

##### Example:

Listening to a song or a story.

Used in mosques for the call to prayer and in schools.



#### 4 Headphones



Headphones give sound to our ears privately.

##### Example:

Listening to a lesson without disturbing others.

Students in the UAE use headphones in online classes.



#### 5 Projector



A projector shows the computer's image on a big screen or wall.

##### Example:

Watching a movie or presentation in class.

Used in UAE schools, conference rooms and events.



### Let's Remember!

- ★ Output devices show, play, or print information.
- ★ We can see it on a screen, hear it with sound, or print it on paper.
- ★ Computers have many ways to give us output.

Smart output devices help us in everyday life in the UAE!



### Quick Check

Circle the output device in each row.

- 
- 
- 

### Summary

- Output devices help us get information from the computer.
- We can see it, hear it, or print it.
- Examples: monitor, printer, speakers, headphones, projector and more.

Computers are useful because they can give us information in many ways!



Think Smart!

In the UAE, computers and smart technology help us build a better future. Look around you – output devices are everywhere!



Circle the device that matches each job.

Which device helps you **see a picture**?



Monitor



Printer

Which device helps you **hear music**?



Printer



speakers



Headphones

Which device makes a **page you can hold**?



Printer



speakers



**True (T) or False (F) ?**

- 1. A monitor shows pictures. \_\_\_\_\_**
- 2. A printer makes copies on paper. \_\_\_\_\_**
- 3. A mouse is an output device. \_\_\_\_\_**
- 4. Speakers let us hear sounds. \_\_\_\_\_**

**Match the device to what it does.**

- |                    |                                     |
|--------------------|-------------------------------------|
| <b>1. Monitor</b>  | <b>● Make copies on papers.</b>     |
| <b>2. Printer</b>  | <b>● lets us hear sound.</b>        |
| <b>3. Speakers</b> | <b>● shows pictures and videos.</b> |

**Circle the output devices.**

**○ Monitor    ○ Keyboard    ○ Printer    ○ Mouse    ○ Speakers    ○ Webcam**

# Lesson 5: Everyday Devices Controlled by Computers

## Learning Objective

### 1CS.05 Know that there are many everyday devices that use computers.



#### Learning Objective

1CS.05 Know that there are many everyday devices that use computers.



In the UAE, we use smart devices and computers in many places every day!

Many devices we use every day have a small computer inside them. These computers help the device work, follow instructions, and make our lives easier and more fun!

The UAE uses technology and smart solutions to help people and build a better future!



#### Smart TV



Computers help smart TVs show videos, play games, and connect to the internet.

##### In the UAE:

We watch educational and cultural programmes about the UAE on smart TVs.



#### Washing Machine



Computers control how the washing machine washes and spins your clothes.

##### In the UAE:

Smart washing machines help save water and energy – important in our desert climate.



#### Microwave Oven



Computers decide how long to heat your food and how much power to use.

##### In the UAE:

Microwaves help us heat food quickly after school or after Iftar during Ramadan.



#### Automatic Door



Computers sense when you are near and open the door automatically.

##### In the UAE:

Automatic doors in malls, metro stations and hospitals make places easier and safer for all.



#### Traffic Lights



Computers control the lights to keep traffic moving safely.

##### In the UAE:

Smart traffic lights in Dubai and Abu Dhabi help reduce traffic and keep roads safe.



#### Elevator (Lift)



Computers help the elevator know which floor to go to.

##### In the UAE:

Tall buildings like the Burj Khalifa use smart elevators to move people quickly.



#### Remember!

- ★ These devices may look different, but they all have a computer inside that helps them work.
- ★ Computers are in many places around us!



#### ★ Quick Check

Circle the devices that use computers.



#### Think!

Can you think of any other devices that use computers at home or at school in the UAE?



Computers are all around us, helping many devices every day!



Proud to be in the UAE!





## Circle the Correct Picture

Which of these has a computer inside?

- Teddy bear 
- Microwave 
- Bicycle 

## Match the Pictures

Match the device to what the computer helps it do:

- |                                                                                                     |                                                                                                                   |
|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| •  Traffic lights  | •  Show videos                   |
| •  Washing machine | •  Controls washing and spinning |
| •  Smart TV        | •  Controls the traffic lights   |

Tick (✓) the correct answer.

**A microwave can use a computer to:**

☐ heat food ☐ wash clothes

**Traffic lights use computers to:**

☐ control the lights ☐ play music

**An automatic door can use a computer to:**

☐ open the door when someone is near ☐ make toast

**A lift can use a computer to:**

☐ choose which floor to go to ☐ wash clothe

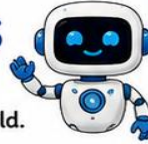
## Lesson 6: Robots Around Us

### Learning Objective

1CS.06 Identify what robots are and where they may be found in the real world

#### Lesson 6

## Robots Around Us



**Learning Objective:** 1CS.06

Identify what robots are and where they may be found in the real world.

### What is a Robot?

A robot is a machine that can follow instructions and do a job.



#### Robots can:

- Sense things around them.
- Follow instructions.
- Move from one place to another.
- Do useful jobs.
- Use electricity or batteries.

### Where Can We Find Robots?

Robots can be found in many places around us.

#### At Home



Robot vacuum cleans the floor.



Robot lawn mower cuts the grass.

#### In Hospitals



Some robots help doctors and nurses with different jobs.

#### In Factories



Robots help make and move products.

#### In Shops and Malls



Some robots help people find places or products.

### Robots in the UAE

#### Delivery Robots in Dubai

Delivery robots can move along pavements, find their way, carry deliveries, and use technology to help them work safely.



#### Robots and Dubai Police

Dubai Police use robots for different services. For example, the robot "Amal" helps educate children during awareness activities.



★ The UAE uses robots and smart technology to help people and make life easier.

### Robot or Not?

Not every machine is a robot.

#### Robot vacuum



✓ Yes

It can move and follow instructions.

#### TV



✗ No

It follows commands but does not move around and perform a job.

#### Remote-control robot car



✓ Yes

It can move according to instructions.

#### Teddy bear



✗ No

A normal teddy bear does not follow computer instructions.

#### Delivery robot



✓ Yes

It can move and deliver items.

### Think and Talk



Have you ever seen a robot?

- Where did you see it?
- What did the robot do?
- Did it move?
- How did it know what to do?



Imagine you are in a mall in Dubai. What could a robot help you do?



Find a shop



Find a product



Give directions



Deliver something



Clean the floor

### ★ Quick Check

1. What is a robot?

- ✓ A machine that can follow instructions and do a job.
- B. A toy that cannot move.

2. Which one is a robot?



A



B

3. Where might we find robots?

- A. At home
- B. In hospitals
- C. In factories
- D. All of these ✓

4. What can a robot do?

- A. Only play games.
- ✓ B. Move, sense things, and follow instructions.



### Key Message

Robots are machines that can follow instructions and do useful jobs. We can find robots at home, in hospitals, factories, shops, and even in places around the UAE! 🤖 🇦🇪





# Unit 2: Computational Thinking

## Lesson 1: Understanding Algorithms

### Learning Objective

1CT.01 Follow the steps in algorithms for everyday tasks.

By the end of the lesson, students will be able to:

- Understand that an **algorithm** is a **set of instruction for a task**.
- Follow steps in the **correct order**.
- Use algorithms for familiar everyday activities.

Year 1 Computing

**Follow the Steps in Algorithms** for Everyday Tasks

UAE Culture Connection

An algorithm is a **set of steps** we follow in the **correct order** to do something.

In the UAE, we follow steps in many important activities. For example, when we prepare for **National Day** celebrations at school.

**Example: Preparing for National Day at School**

**1**  
First, wear your UAE themed clothes.

**2**  
Next, take your UAE flag.

**3**  
Then, pack your school bag.

**4**  
After that, go to school.

**5**  
Finally, join the celebration and have fun!

**Why do we follow the order?**

If we change the order, the task may not work!

When we follow the steps in the right order, everything works well!

**We follow algorithms every day!**

Brushing teeth

Eating breakfast

Getting ready for school

Doing homework

**Remember!**

An algorithm has steps.

We follow the steps in the correct order.

Algorithms help us do things the right way!

**Activity**

Look at the pictures below. Cut and paste the steps in the correct order for making Arabic Coffee (Gahwa). Write 1, 2, 3, 4 in the boxes.

Add cardamom.

Add the coffee.

Pour the coffee into the cups.

Boil the water.

**I can...**

I can follow steps.

I can put steps in order.

I understand what an algorithm is.

Algorithms help us every day in the UAE and everywhere!



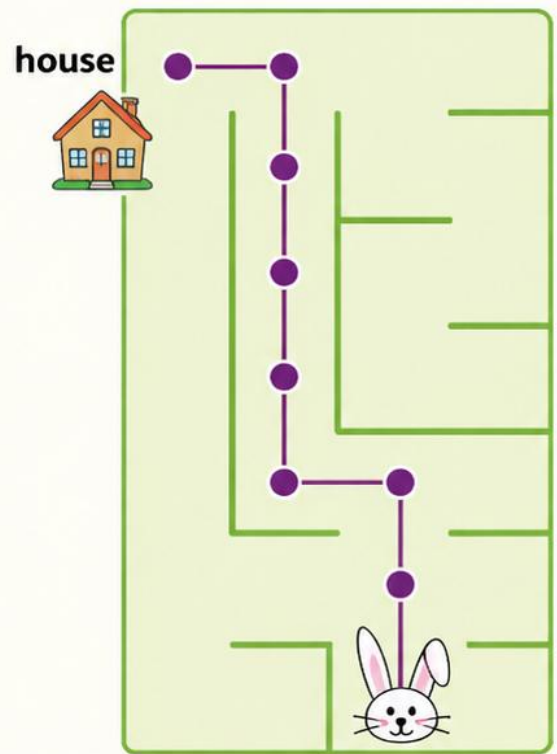
**Instruction:** information about how something should be done.

**Task:** an activity.

### Algorithm for solving a problem

The instructions below are for a maze.

The instructions tell the rabbit how to get to the house.

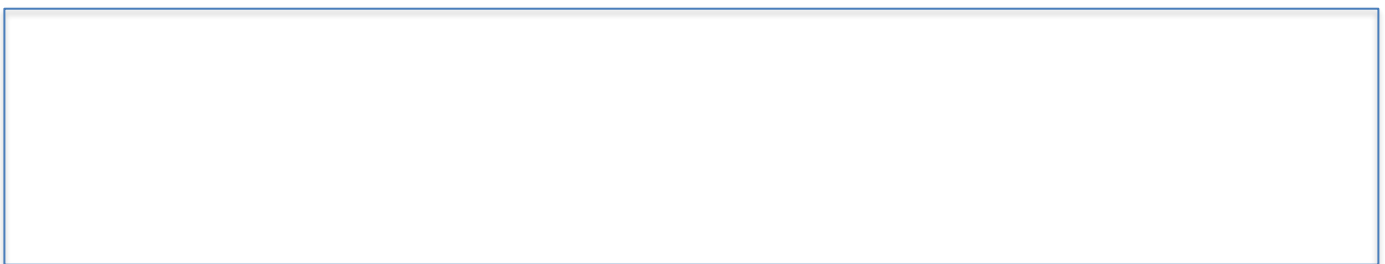


| Step | Instruction      |                                                                                     |   |   |
|------|------------------|-------------------------------------------------------------------------------------|---|---|
| 1    | Move 2 steps up  |   | 2 | ↑ |
| 2    | Move 1 step left |  | 1 | ← |
| 3    | Move 4 steps up  |  | 4 | ↑ |
| 4    | Move 1 step left |  | 1 | ← |

## Lesson 2:

### Learning Objective

1CT.02 Identify single errors in algorithms that represent everyday events or tasks.





1CT.03 Know how to give simple instructions, using directional language (forward, backwards, left, right), to navigate a path.

1CT.04 Suggest sets of ordered instructions to complete simple tasks, such as drawing a picture of a particular object or building a brick tower.

1CT.05 Know that an algorithm is a set of instructions to complete a task or to solve a problem.

1CT.06 Know that the order of instructions is important when creating algorithms.

1CT.07 Suggest ways that algorithms can be changed to affect the outcome.